Writing a Bachelor Thesis in Applied Economics

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Roadmap for today

Session 1: From Question to Feasible Design

- What makes an applied thesis good?
- Crafting a research question (RQ)
- Reading papers efficiently (hands-on)
- Data availability & quality
- Emprical Research Methods

Session 2: From Analysis to Write-up

- Reading empirical tables
- Interpretation
- Robustness checks
- Falsification tests
- Introduction, abstract, discussion

Communication Policy

- Questions are welcome!
- Remember, this session is for you!
- **For research-related questions, contact your advisor first.**

Why this structure?

It mirrors the real research process.

1. **Start with a question** What do you want to understand or explain?
2. **Check feasibility** Is there data? Can this be answered empirically?
3. **Read the literature** What do we already know? How have others approached this?
4. **Design your empirical strategy** Identification, variables, and methods
5. **Collect and analyze data** Run regressions, create tables, test hypotheses
6. **Validate your results** Robustness checks, falsification tests
7. **Interpret findings** What do the results actually mean?
8. **Write the paper (last!)** Introduction, abstract, and discussion come *after* you know the results

What Makes an Applied Economics Thesis "Good"?

- **Clear, narrow but interesting question** that is answerable with available data
- **Feasible, transparent design**: research method that suits the context
- **Credible evidence**: careful description, sound comparisons, appropriate controls
- **Economic interpretation**: magnitudes, units, and uncertainty
- **Readable communication**: clean tables/figures; concise writing

The Research Question

Finding Ideas

- **Motivate Research with Real-World Events:**
  - Identify policy debates, economic crises, or industry trends
  - Use news, reports, and class topics as inspiration
  - Ask: "Which economic problems lack clear answers?"

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  - *Journal of Economic Literature*, *Economic Perspectives*
  - Track citations forward/backward to follow papers
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- **Apply New Methods to Economics Questions:**
  - Introduce data science, machine learning, or modern econometric techniques
  - Revisit classical questions with new tools
  - Combine approaches from different fields for fresh insights

Research Questions: Types

- **Descriptive**: What is happening? (patterns, levels, trends)
- **Predictive**: What will happen? (forecasts)
- **Causal**: What is the effect of X on Y? (counterfactual)

Examples of Research Questions That Are Too Broad

- How does *education* affect *labor markets*?
- What is the impact of *immigration* on *the economy*?
- How do *taxes* influence *firm behavior*?
- What drives *inequality*?
- How do *environmental policies* shape *growth*?
- How does *digitalization* affect *productivity*?

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Why too broad? Missing unit, time, setting, measurable outcome, and feasible data link.

Sharpened and Feasible Versions (Examples totally made up!)

- Education → *Did the 2014 Swiss vocational reform change starting wages of apprentices in Canton Zurich?*
- Immigration → *Did net immigration 2010–2020 raise median rents more in urban than rural municipalities?*
- Taxes → *How did the 2019 corporate tax reform affect investment among medium-sized firms?*
- Inequality → *Did expansion of childcare subsidies 2012–2018 reduce the gender pay gap among new parents?*
- Environment → *Did the 2015 CO_2 levy increase energy efficiency of manufacturing plants?*
- Digitalization → *Did broadband roll-out 2008–2015 raise total factor productivity of SMEs in Canton St. Gallen?*

Reading Papers efficiently & Literature review

How to Read an Economics Paper Efficiently

- Read: abstract $\rightarrow$ intro first paragraphs $\rightarrow$ (figures/tables) $\rightarrow$ conclusion
- Focus on: **question**, **data**, **empirical approach**, **main finding**
- Skim methods for identification idea rather than derivations
- Note limitations and external validity

Reading Papers = Filtering, Not Consuming

Key idea: You are not supposed to read every paper from top to bottom!

- Goal is **finding relevant papers quickly**
- Most papers are **not directly useful** for your question
- Goal: decide early $\rightarrow$ **continue or stop reading**

Typical workflow

1. Skim abstract + first paragraphs of intro → Is this related to my question?
2. Look at tables/figures → What do they actually find?
3. **Decision point:**
   - Not relevant → **stop reading**
   - Relevant → continue with intro + conclusion
4. Only then: dive deeper into data and methods (if needed)

Why this works: Economics papers are written so the introduction already summarizes question, data, method, and previews results.

How to Read a Paper for Your Thesis

Focus on what you can reuse:

- **Research question:** how it is motivated and formulated
- **Data:** unit, time period, variables, cleaning steps
- **Identification:** the core logic (in words, not math)
- **Main table:** how the specification develops across columns
- **Interpretation style:** how magnitudes are explained

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A simple note-taking template

1. What is the question?
2. What data are used? (unit, years, variation)
3. What is the empirical approach?
4. What is the main result (in units)?
5. How do they check robustness?

Exercise

Bertrand, Marianne, and Sendhil Mullainathan. 2004. "Are Emily and Greg More Employable Than Lakisha and Jamal? A Field Experiment on Labor Market Discrimination." American Economic Review 94 (4): 991–1013.

https://pubs.aeaweb.org/doi/pdfplus/10.1257/0002828042002561

- Please identify: question, data source, unit and years, empirical approach, main result by:
  - reading the abstract
  - reading the introduction
  - reading the conclusion

Literature Review

A good literature review should:

- be organized around and focused on your research question
- synthesize existing work into a summary of what is and is not known
- identify areas of controversy or open questions in the literature

It should not:

- list everything that you have read, even if unrelated.
- present a sequence of summaries for individual papers.
- add spurious citations that you have not personally read

From List to Synthesis: Mini-Example

Unhelpful list

Smith (2018) studies minimum wages and teen employment and finds negative effects. Lopez (2019) examines sector-specific outcomes and reports mixed results. Chen (2021) looks at low-wage workers and notes heterogeneous effects. Miller (2022) studies overall employment and finds no significant change. Cole (2015) mentions minimum wage, but it seems to be unrelated to my topic.

From List to Synthesis: Mini-Example

Better synthesis (idea-focused)

The literature on minimum wage effects covers several distinct worker groups. Research on **teen employment**, such as Meyer et al. (2015) and Smith (2018), typically finds negative or modestly negative effects. Studies focusing on specific **low-wage sectors**, for example Lopez (2019), report mixed evidence that is often sensitive to local labor-market conditions. Work examining **individuals earning low wages prior to policy changes**, including Chen (2021), documents heterogeneous impacts depending on program intensity and worker characteristics. **Despite substantial evidence for these subgroups, credible estimates of the impact on overall employment remain scarce, which this thesis aims to address.**

Data

Data: Availability and Quality

- Access: public vs licensed; replication packages
- Coverage: time, geography, population; unit of observation
- Measurement: definitions, consistency, missingness, outliers
- Sample size and comparability across groups/periods

Where to Find Data in Practice

Start with existing sources before designing your project

- **Replication packages**
  - Published papers (AER, QJE, etc.)
  - Often include cleaned data + code
- **Public data portals**
  - World Bank, OECD, Eurostat
  - National statistical offices
- **Administrative / micro data**
  - Surveys (e.g., labor force, household)
  - Firm or tax data (often restricted access)
- **Other sources**
  - Company reports, web scraping, APIs
  - Experiments or own data collection (rare for theses)

If you cannot clearly identify a dataset, your research question is not yet feasible.

A Practical Rule: Start with Data

- Many theses fail because of **no usable data**
  - If you find an interesting dataset
  - Then develop a research question around it
- Ask yourself early:
  - Can I access the data?
  - Is it at the right level (individual, firm, region)?
  - Does it vary enough to answer my question?

No data $\Rightarrow$ no thesis (at least in empirical economics).

Empirical Research Methods

Map Your Question to a Simple Empirical Approach

You have to find a reasearch design that is suitable to answer the research question with the given data!

- Relationship ("How are X and Y related?") $\rightarrow$ correlation, scatter plot, OLS
- Group differences (A vs B) $\rightarrow$ means, t-test, OLS with dummies
- Event/policy timing $\rightarrow$ pre/post comparison; simple DiD logic
- Heterogeneity $\rightarrow$ subgroup analysis; interactions in OLS, causal Machine Learning
- Surveys $\rightarrow$ descriptive stats; cross-tabs; weighted means

Method Choice and Identification (Overview)

Setting	Go-to design (key assumption)
Descriptive patterns	Descriptives, OLS as description (no causal claim)
Policy with clear timing	Difference-in-Differences or event study (parallel trends)
Unobservables drive endogeneity	Instrumental variables (relevance, exclusion)
Threshold assignment	Regression discontinuity (continuity, no sorting)
Selection on observables	Matching or weighting (overlap)
Panels with heterogeneity	Fixed effects (strict exogeneity)

Every setting calls for another empirical method.

So choose the right method for the questions you want to answer!

Presenting and Reading Empirical Findings

Reading Empirical Tables: Quick Guide

- Start with the Title + Notes
- Look at **coefficient and standard error**: sign, magnitude, precision
- Read the **notes**: sample, units, controls, fixed effects, clustering
- Focus on **economic magnitude**: translate to meaningful units

Example from a Published Paper

TABLE 1—MEAN CALLBACK RATES BY RACIAL SOUNDINGNESS OF NAMES

	Percent callback for White names	Percent callback for African-American names	Ratio	Percent difference (p-value)
Sample:				
All sent resumes	9.65	6.45	1.50	3.20
	[2,435]	[2,435]		(0.0000)
Chicago	8.06	5.40	1.49	2.66
	[1,352]	[1,352]		(0.0057)
Boston	11.63	7.76	1.50	4.05
	[1,083]	[1,083]		(0.0023)
Females	9.89	6.63	1.49	3.26
	[1,860]	[1,886]		(0.0003)
Females in administrative jobs	10.46	6.55	1.60	3.91
	[1,358]	[1,359]		(0.0003)
Females in sales jobs	8.37	6.83	1.22	1.54
	[502]	[527]		(0.3523)
Males	8.87	5.83	1.52	3.04
	[575]	[549]		(0.0513)

Notes: The table reports, for the entire sample and different subsamples of sent resumes, the callback rates for applicants with a White-sounding name (column 1) an an African-American-sounding name (column 2), as well as the ratio (column 3) and difference (column 4) of these callback rates. In brackets in each cell is the number of resumes sent in that cell. Column 4 also reports the p-value for a test of proportion testing the null hypothesis that the callback rates are equal across racial groups.

Source: Bertrand and Mullainathan (2004)

Interpreting Results

- Translate coefficients into **units**: e.g., "0.10 means a 10 percentage point increase"
- Compare to **baseline means** to gauge size
- Discuss **direction, size, and uncertainty** (CI or SE)
- **Mechanisms** only if supported by evidence; avoid speculation
- Keep claims aligned with **design limits** (descriptive vs causal)

Exercise

- Look at the table on the slide before. What results do you see?
- Check how the paper describes the results.

Robustness Checks

Why Do We Need Robustness and Falsification Tests?

Core problem: Empirical results can be misleading

- Results may depend on **arbitrary choices**
  - Functional form, controls, sample, specification
- Apparent effects may be driven by **confounding factors**
  - Omitted variables, trends, selection
- Some findings can arise by **chance or data mining**

What we want: Credible (causal) interpretation

Why Do We Need Robustness and Falsification Tests?

- **Robustness checks**: Does the result hold under reasonable alternative specifications?
- **Falsification tests**: Do we find "effects" where there should be none?

Interpretation:

- If results survive $\rightarrow$ more credible
- If results disappear $\rightarrow$ likely not causal

Key idea: You are trying to break your own result. If you cannot, it becomes more convincing.

Robustness Checks

- Alternative functional forms; transformations
- Different sets of controls; drop/keep influential observations
- Alternative clustering of SEs; heteroskedasticity-robust SEs
- For panels: add time/region fixed effects; different sample windows
- For RDD: bandwidth and polynomial sensitivity; placebo cutoffs
- For DiD: add group-specific trends; test sensitivity to cohort definitions

It is usually a good idea to get inspiration for robustness check in the literature!

Falsification Tests

- **Placebo timing**: apply the treatment date to pre-periods
- **Placebo outcomes**: choose outcomes the treatment should not affect
- **Pre-trends**: check parallel trends before the policy/event
- **No-effect groups**: groups unlikely to be affected (e.g., ineligible units)
- **Permutation or randomization inference** (when applicable)

Writing

Typical Structure of an Economics Thesis

1. **Introduction:**
   - Introduces the topic and motivates the research question
2. **Literature Review:**
   - Summarizes relevant research
   - Highlights how your work contributes to the field
3. **Data:**
   - Describes data sources, variables, and sample
4. **Methods/Approach:**
   - Explains theoretical framework, empirical strategy, or modeling approach
5. **Results/Findings:**
   - Presents your main results clearly (tables, figures, or models)
   - Includes robustness checks or sensitivity analyses
6. **Discussion & Conclusion:**
   - Interprets results and relates them back to the research question
   - Suggests implications, limitations, or directions for future research

Writing the Abstract (120–150 words)

- Research question in one sentence
- Data: source, unit, N, years
- Method: design and identification
- Key result with magnitude and uncertainty
- Implication (no citations; minimal jargon)

Thesis Structure — Details (Part 1 of 3)

1. Introduction

- Motivation: why the question matters
- Research question
- Contribution: what you add to the literature
- Short preview: data, method, headline finding
- Roadmap (1–2 sentences)

2. Literature Review

- Organize by *idea/approach*, not by author
- Synthesize patterns; identify the gap
- Position your thesis (how you differ/extend)
- Keep focus: 3–6 core papers

Thesis Structure — Details (Part 2 of 3)

3. Data

- Source(s), unit of observation, period
- Variable construction: outcome, regressors, controls
- Sample selection rules; missing data handling
- Descriptive statistics (levels, variation)

4. Methods / Empirical Strategy

- Identification idea in words; key assumptions
- Estimation approach (e.g., OLS, DiD, IV, RDD, FE)
- Standard errors (e.g., clustering) and diagnostics

Thesis Structure — Details (Part 3 of 3)

5. Results

- Main specification table
- Interpret magnitudes in real units
- Robustness checks (most informative only)
- Falsification tests (placebos, pre-trends, etc.)

6. Discussion and Conclusion

- Recap the main finding (1–2 sentences)
- **Answer to your research question**
- Mechanisms or plausible channels (when supported)
- Threats to validity and how you addressed them
- External validity and policy relevance
- What remains uncertain and future work

Final Note

Keep it feasible, credible, and clear. You can do this!

Questions?

Questions?

Are there any uncertainties/ problems/ specific questions you have?

What's next?

What's next?

Two discussion meetings:

- Friday, 10.04.2026, 08:15–12:00
- Tuesday, 21.04.2026, 14:15–18:00

You present your ideas, we (as a group) discuss and help to refine them.

Please send me the proposal via email until one week before the presentation (03.04./14.04.).
I will share them with you, so you can all read the proposals and prepare your discussion.

Schedule Discussion Session

Discussion Session 1: Friday, 10.04.2026 — 08:15–12:00 in 01-308

Presenter	Discussant
Davide Vaz Sequeira	Lennart Herlyn
Mathieu Fauchère	Davide Vaz Sequeira
Lennart Herlyn	Teva Janura
Teva Janura	Mathieu Fauchère

Discussion Session 2: Tuesday, 21.04.2026 — 14:15–18:00 in 07-001

Name	Discussant
Paul Ludmann	Grace Yan
Grace Yan	Philipp Nietgen
Charlotte De Jong	Paul Ludmann
Philipp Nietgen	Charlotte De Jong

Reminder Deliverables

1. **Research Proposal (25%):**
   - Submit an extended research proposal (max. 10 pages)
   - Sent it via email at least **seven days** before the workshop.
2. **Presentation (50%):**
   - Prepare slides to present your research idea briefly in class (15 minutes)
   - Serve as a discussant for other participants' proposals, 5 min each
3. **Active Participation (25%)** in at least two sessions:
   - Carefully read proposals in advance, take notes
   - Actively contribute to discussions
   - Missing a session leads to a fail, so plan ahead!